MILITARY SPECIFICATION

DELAY ELEMENT, LOADING, ASSEMBLING AND PACKING

This specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements for the loading, assembling and packing and Quality Assurance Provisions for one type of delay element assembly used in the XM87 Canister Assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. - The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specificastions and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

MILITARY

MIL-P-14232	Parts, Equipment and Tools for Army Material, Packaging and Packing of
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods
MIL-STD-650	Explosives, Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9384909	Assembly, Delay Element
9384879	Pin, Firing
9384895	Housing, Upper
9384896	Housing, Lower
9384908	Delay Cord Assembly

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

CODE OF FEDERAL REGULATIONS

Title 49	Transportation, Parts 100-199

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(The Code of Federal Regulation is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)".

2.3 Order of precedence. - In the event of conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Delay element assembly. - The Delay Element Assembly (DEA) shall comply with all requirements specified on Drawing 9384909 and with all requirements specified therein.

3.3 Moisture content.

3.3.1 Lead styphnate. - The moisture content of the lead styphnate at the time of loading the DEA shall not exceed 0.15 percent when tested as specified in 4.5.2.

3.3.2 NOL-130. - The moisture content of the NOL-130 at the time of loading the primer shall not exceed 0.15 percent when tested as specified in 4.5.2.

3.3.3 Delay cord. - The moisture content of the delay cord core material shall not exceed TBD percent when tested as specified in 4.5.2.

3.4 Performance.

3.4.1 Sensitivity. - The primer shall initiate when impacted with 1.20 $\pm$ 0.05 inch-ounces of energy applied with a firing pin tip conforming to drawing 9384879.

3.4.2 Output. - The time elapsed between primer initiation and delay cord output shall be from 120 to 200 seconds.

3.5 Environmental conditioned.

3.5.1 High temperature. - The DEA shall comply with the requirements of 3.4 while being exposed to the hot temperature environment as specified in 4.5.4.2.

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3.5.2 Low temperature. - The DEA shall comply with the requirements of 3.4 while being exposed to the low temperature environment as specified in 4.5.4.3.

3.5.3 Thermal shock. - The DEA shall comply with the requirements of 3.4 at room temperature following the thermal shock exposure as specified in 4.5.4.4.

3.5.4 Operational vibration. - The DEA shall comply with the requirements of 3.4 at room temperature following the vibration exposure specified in 4.5.4.5.

3.5.5 Mechanical shock. - The DEA shall comply with the requirements of 3.4 at room temperature following the shock exposure specified in 4.5.4.6.

3.5.6 Leadage (immersion). - The DEA shall comply with the requirements of 3.4 while being subjected to the leakage environment specified in 4.5.4.7.

3.6 First article inspection. - This specification makes provision for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

3.7 Workmanship. - All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be clean and free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, oil, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specfication shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provision of 4.3.2 and 4.3.3. The first article sample shall consist of items and sample quantities as in Table I.

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078 and as specified herein.

4.3.4 Requalification. Canister assembly shall be subjected to another First Article Inspection in accordance with Table I, if any major changes are made which affect the form, fit, geometry and functioning of the Delay Element Assembly as defined in paragraph 3.2.3.

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Table I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-D-70669(AR)

PARAGRAPH	TITLE: Delay Element Assembly		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY: ---
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Cord Assembly (Dwg. 9384908)				
	Examination for Defects	50	50-0-1	3.2	4.4.2.1
	Stab primer sensitivity		50-0-2	3.2	4.5.4.1
	Low Housing (Dwg. 9384896)				
	Examination for Defect	50	50-0-1	3.2	4.4.2.2
	Delay Element Assembly (Dwg. 9384909)				
	Examination for Defect	120	120-0-1	3.2	4.4.2.3
	Function Tests				
	High Temperature		20-0-1	3.2	4.5.4.2
	Low Temperature		20-0-1	3.2	4.5.4.3
	Thermal Shock		20-0-1	3.2	4.5.4.4
	Operational Vibration		20-0-1	3.2	4.5.4.5
	Mechanical Shock		20-0-1	3.2	4.5.4.6
	Leakage		20-0-1	3.2	4.5.4.7

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of DEA's shall contain:

a. Each explosive material from not more than one lot.

b. Delay cord from not more than one lot.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Level II, Normal, Single Sample Inspection.

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MIL-D-70699(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Delay Cord Assembly				9384908
					NEXT HIGHER ASSEMBLY 9384909
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Explosive Present on Exterior Surface		100%	3.2	Visual
Major					
101	Output End Diameter improper		.40%	3.2	Gage
102	Output End Length improper		.40%	3.2	Gage
103	Input End Senstivity Diameter improper		.40%	3.2	Gage
104	Moisture Content of Explosives improper		Each lot	3.3	4.5.2
105	Sealant Missing from Crimp Area		.40%	3.2	Visual
Minor					
201	Poor Workmanship		.65%	3.7	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70699(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.2	Lower Housing	1	1	9384896
				NEXT HIGHER ASSEMBLY: 9384909

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Longer side of slider cavity improper		.40%	3.2	Gage
102	Shorter side of slider cavity improper		.40%	3.2	Gage
103	Width of slider cavity improper		.40%	3.2	Gage
104	Barrier Tab missing		.40%	3.2	Visual
105	Location of Cord Input End Cavity Face improper		.40%	3.2	Gage
106	Location of Initiator Alignment Hole improper		.40%	3.2	Gage
107	Size of Initiator Alignment Hole improper		.40%	3.2	Gage
108	Surface A not sandblasted		.40%	3.2	Visual
Minor					
201	Poor Workmanship		.65%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-D-70699(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Delay Element Assembly		9384909
			NEXT HIGHER ASSEMBLY ---

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Delay cord assembly not flush		.40%	3.2	Gage
102	Protrusion of stab primer incorrect		.40%	3.2	Gage
103	Sealant missing from joints		.40%	3.2	Visual
104	Encapsulation improper		.40%	3.2	Visual
105	Sealant applied incorrect		.40%	3.2	Visual
106	Cup below minimum distance permitted		.40%	3.2	Gage
107	Functional test			3.4	4.5.4
	Lot size				
	1201 - 10,000 — 200-1-2				
	Hot (145°F) — 75				
	Ambient (70±10°F) — 50				
	Cold (-40-5°F) — 75				
	10,000 - 35,000 — 315-2-3				
	Hot (145°F) — 105				
	Ambient (70±10°F) — 105				
	Cold (-40-5°F) — 105				
Minor					
201	Poor Workmanship		.65%	3.7	Visual

NOTES:

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4.5 Methods of inspection.

4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.2 Moisture content of explosive material. - The moisture content of explosive material shall be determined at the time of loading. The method shall be in accordance with MIL-STD-650, Method 101.5. A two gram sample shall be selected from that quantity to be used to produce a lot and shall meet the requirements of the applicable drawing. The sample disk shall be covered with a sheet of polytetrafluoroethylene. If the moisture content fails to comply with specified requirement of 3.3, the batch shall be rejected.

4.5.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Methods" column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and section 6.3 herein.

4.5.4 Functioning tests. - For all functional tests, the DEA will be securely attached to a flat plate simulating the mine interface. Failure of the interface shall not be considered a DEA failure. A cover shall be installed prior to testing simulating the 9384895 upper housing as shown on 9384912. The DEA's will be tested as follows.

4.5.4.1 Function. - Operation of the DEA in conjunction with exposure to an environment as specified in the remainder of section 4.5.4, shall consist of the following:

a. Stab the primer with 1.20 ± .05 inch ounce energy (.15 ± .01 ounce weight dropped 8.0 ± .1 inch or other combinations) using firing pin tip identical to 9384879.

b. Time the interval between primer initiation and output firing of delay cord. Timeout shall be within range specified in 3.4.2.

Post fire inspection shall show no evidence of delay cord metal sheath material within the slider cavity.

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4.5.4.2 High temperature function. - Twenty units will be tested per MIL-STD-810, Method 501.1, Procedure I (48 hours at +160°F). Per Step 4 of Method 501.1, stabilize the unit at +145°F and function per 4.5.4.1 while the temperature is maintained at +145°F.

4.5.4.3 Low temperature function. - Twenty units will be tested per MIL-STD-810, Method 502.1 except the storage temperature will be -65°F for 24 hours. Per Step 4 of Method 502.1, stabilize the unit at -40°F and function per 4.6.4.1 while the temperature is maintained at -40°F.

4.5.4.4 Thermal shock. - Twenty units will be tested per MIL-STD-810, Method 503.1 (3 thermal shock cycles during a 24 hour period). Per Step 9 of Method 503.1 function per 4.5.4.1 at room temperature.

4.5.4.5 Operational vibration. - Twenty units will be tested per MIL-STD-810,Method 514.3, Category 1, Prodedure 1, Figure 514.3-7 (1.98G rms, vertical), Figure 514.3-8 (2.00G rms, transverse), and Figure 514.3-9 (2.54G rms, longitudinal). Test duration will be 120 minutes for each axis. The axis through the DEA (perpendicular to the flat(er) faces) will be tested as the transverse axis. The longitudinal axis lies at right angles with the .81 inch long slider cutout in the center of the housing. The vertical axis is perpendicular to this slot. Following this test the units will be functioned per 4.5.4.1.

4.5.4.6 Mechanical shock. - Twenty units will be tested per MIL-STD-810, Method 516.3, Procedure I, functional shock for all ground equipment (40 g's). Following this test the units will be functioned per 4.5.4.1.

4.5.4.7 Leakage (immersion). - Twenty units will be tested per MIL-STD-810, Method 512.1, Procedure I, except the unit should be functioned per 4.5.4.1 and immediately (within 3 seconds after initiation) shall be immersed in the water. Time out shall occur with the unit remaining immersed.

4.6 Check test for possible deterioration. - The following test is designed to prevent DEA's that have deteriorated from being assembled into major ammunition items. The test shall be performed on DEA's from each lot which have been stored under normal conditions for more than two (2) years or under adverse conditions for any period of time or at the request of the Contracting Officer. If the specification for the ammunition item into which DEA's are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting Officer will arrange for the contractor to be reimbursed for expense incurred in the performance of check tests for possible deterioration. DEA's shall be selected from each lot for this test as per 4.6.1.

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4.6.1 Sensitivity and output. - Fifty (50) DEA's shall be selected from each lot for these tests. The test shall be performed as specified in 4.5.4.1 with no prior environmental exposure and using equipment in accordance with 4.5.4. Any DEA that fails to comply with the requirements of 4.5.4.1 shall be classed defective. If one (1) or more DEA's are classed defective, the lot shall be rejected and not used in the loading operation.

5. PACKAGING

5.1 PACKAGING. The DEA shall be packaged in accordance with level B of MIL-P-14232. The DEA's shall be partitioned and cushioned as required.

5.2 Packing. - The DEA shall be packed in accordance with Level B of MIL-P-14232 and Code of Federal Regulations, Title 49, with the following mandatory requirement. The outer packing box shall be wood conforming to class 1, style 4, in accordance with PPP-B-621 or fiberboard conforming to 12H, 23F or 23H of CFR, Title 49.

5.3 Marking. - The marking of the outer box shall be in accordance with MIL-STD-129 and DOT marking in accordance with CFR, Title 49.

5.4 Shipping. When DEA's from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use in M87 canisters.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submissions of inspection equipment designs for approval. - (See MIL-A-48078) Submit equipment designs as required, to COMMANDER: AMSMC-QAT-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the Contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Distribution of AmmunitionData Cards. - Distribution of data cards shall include the following: COMMANDER, ARDEC, ATTN: DRDAR-QAA-Q, Picatinny Arsenal, New Jersey 07806-5000.

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6.6 Drawings. - Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Command (ARDEC) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.7 Subject term (key word) listing.

Delay Element Assembly
Firing Pin
Delay Cord Assembly
Borerider Timer Assembly

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A478)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-70669	DELAY ELEMENT, LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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